ARP Poisoning and Man-In-The-Middle Attack – LAB Report

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ABSTRACT

This lab exercise aimed to carry out an ARP cache poisoning (ARP spoofing) attack in a controlled virtual environment in order to establish a Man-in-the-Middle (MitM) position between a victim (Ubuntu) and the gateway. The tools arpspoof (from the dsniff suite), urlsnarf, and Wireshark were used to execute and validate the attack.

Analysis of the ARP tables before and after the attack confirmed the corruption of the IP-MAC mapping, while the interception of HTTP traffic via urlsnarf demonstrated the attacker's successful interception. The results highlight the vulnerability of the ARP protocol and the need to implement protective measures such as Dynamic ARP Inspection (DAI) and the use of encrypted protocols.

1.INTRODUCTION

The Address Resolution Protocol (ARP) is a procedure that connects an ever-changing Internet Protocol (IP) address to a fixed physical machine address, also known as a Media Access Control (MAC) address, in a local-area network (LAN) (Fortinet, n.d.). ARP is stateless and has no authentication mechanism, which makes it vulnerable to ARP cache poisoning attacks (Yllia & Fejzajb, 2021).

Using ARP spoofing, an attacker can intercept the communication between two users or a user and the gateway, and can control or monitor the transferred data while the victims remain unaware. This is known as a Man-in-the-Middle (MitM) attack (Javeed et al., 2020).

The objectives of the lab exercise were:

- To set up an isolated virtual network with two VMs (Kali Linux as the attacker, Ubuntu as the victim) and a virtual router (gateway).
- To execute an ARP spoofing attack so that the attacker can intercept the data flow.
- To verify the attack by observing changes in the ARP tables and intercepting HTTP traffic using the urlsnarf and Wireshark tools.

2.VIRTUAL BOX SET UP

2.1 Requirements

- VirtualBox (latest version)
- Kali Linux (latest version) as the attacker
- Ubuntu (latest version) as the victim

2.2 Environment

The first step was to create a custom NAT Network in VirtualBox and assign it to both machines (Kali and Ubuntu). The DHCP server was disabled to allow static IP assignment. The NAT Network was configured with the IP range 172.16.0.0/24.

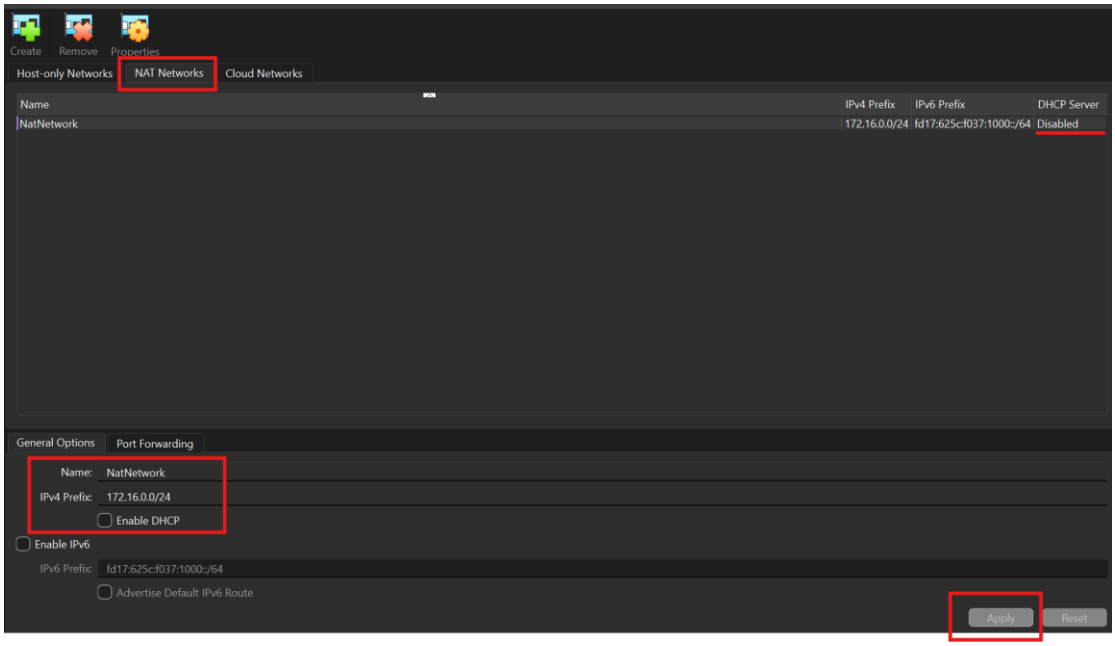


Figure 1: NAT Network configuration.

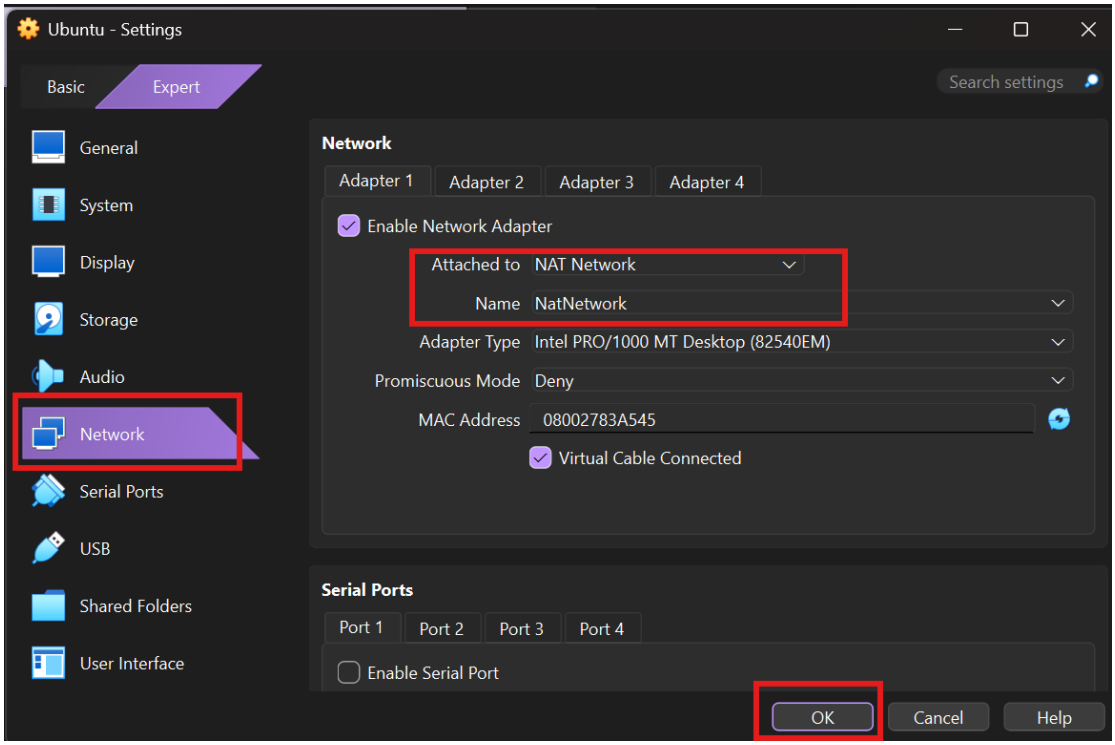


Figure 2: NAT Network assignment.

After starting up both machines, next step is to assign their static IPs and gateway IP addresses. The process is the same for both machines.

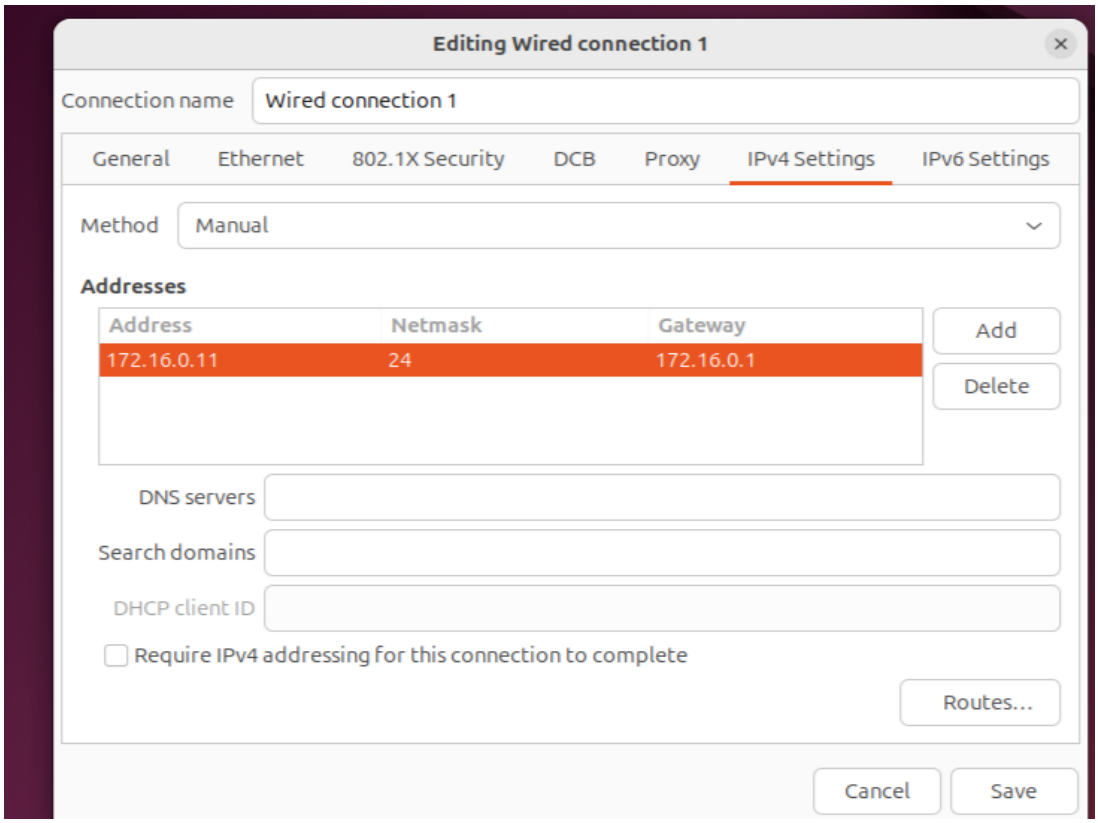


Figure 3: Static IP assignment on Ubuntu

Connectivity was tested using ping commands:

ping <gateway IP> (on Kali&Ubuntu)
ping <Attacker IP> (on Ubuntu)
ping <Victim IP>(on Kali)

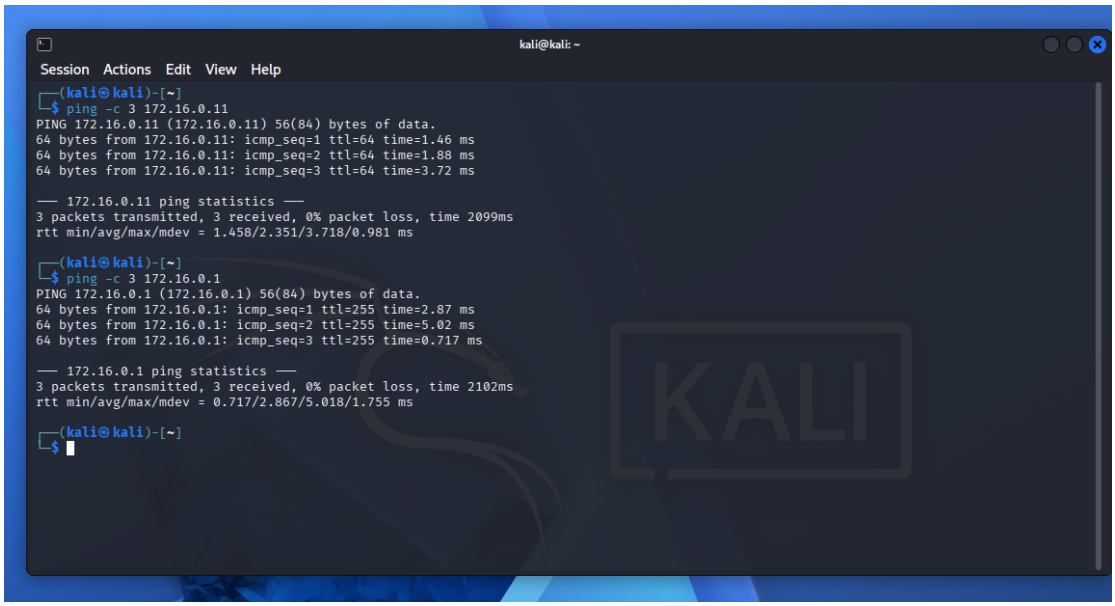


Figure 4: Successful ping from Kali to gateway and Ubuntu

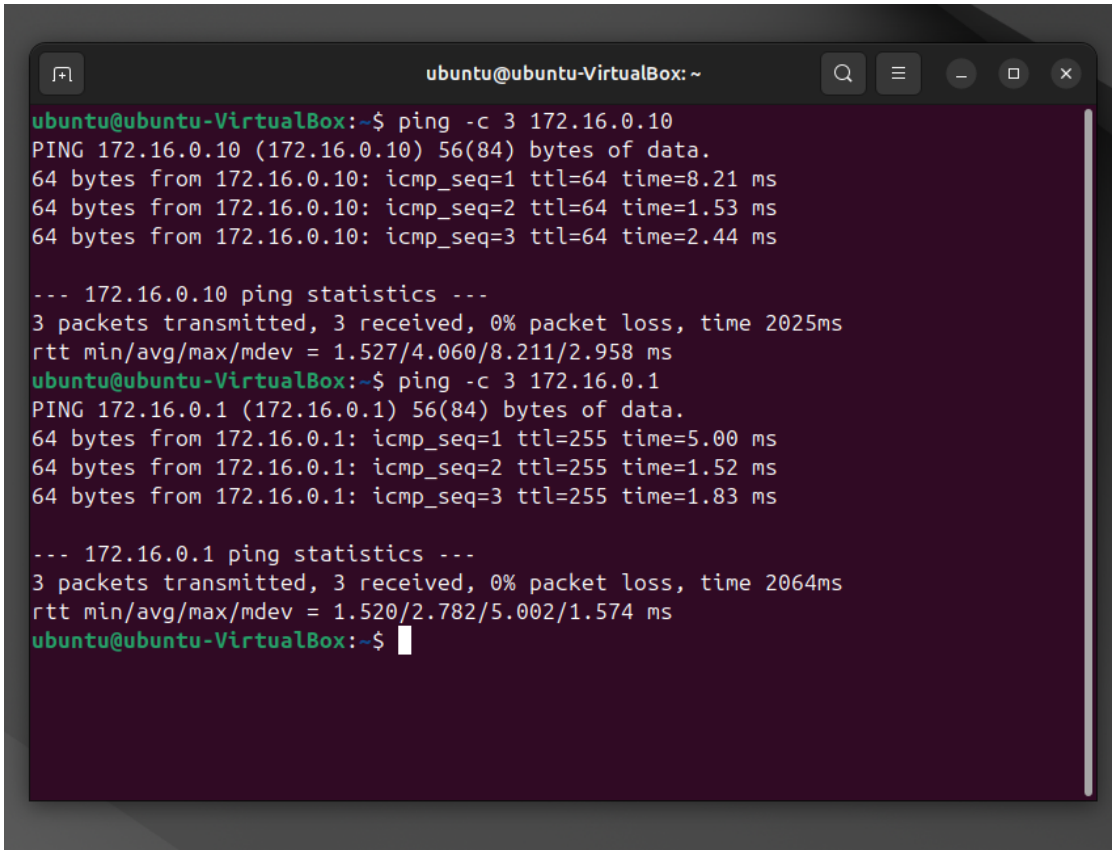


Figure 5: Successful ping from Ubuntu to gateway and Kali

Before the attack, the MAC addresses of all devices were recorded:

- **Kali MAC:** obtained via ip link show eth0 (or eth1)
- **Ubuntu MAC:** obtained via ip link show enp0s8
- **Gateway MAC:** obtained from Ubuntu's ARP table after a ping to 172.16.0.1

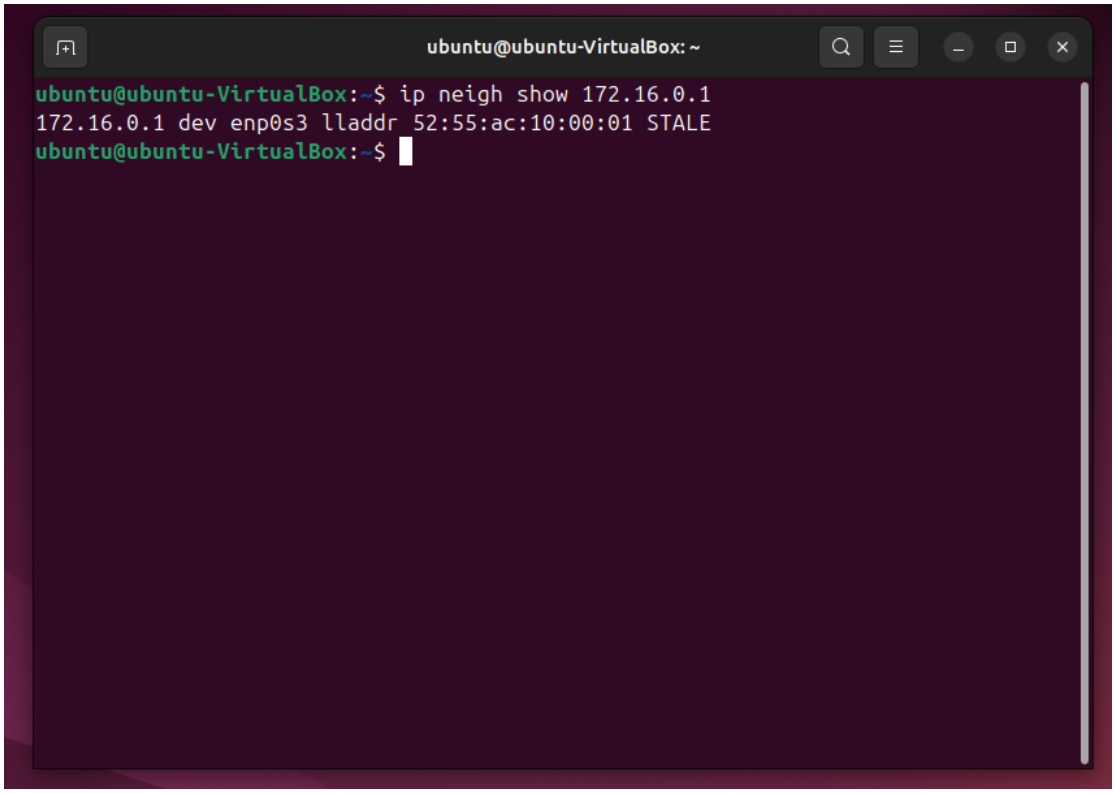


Figure 6: Gateway's MAC address

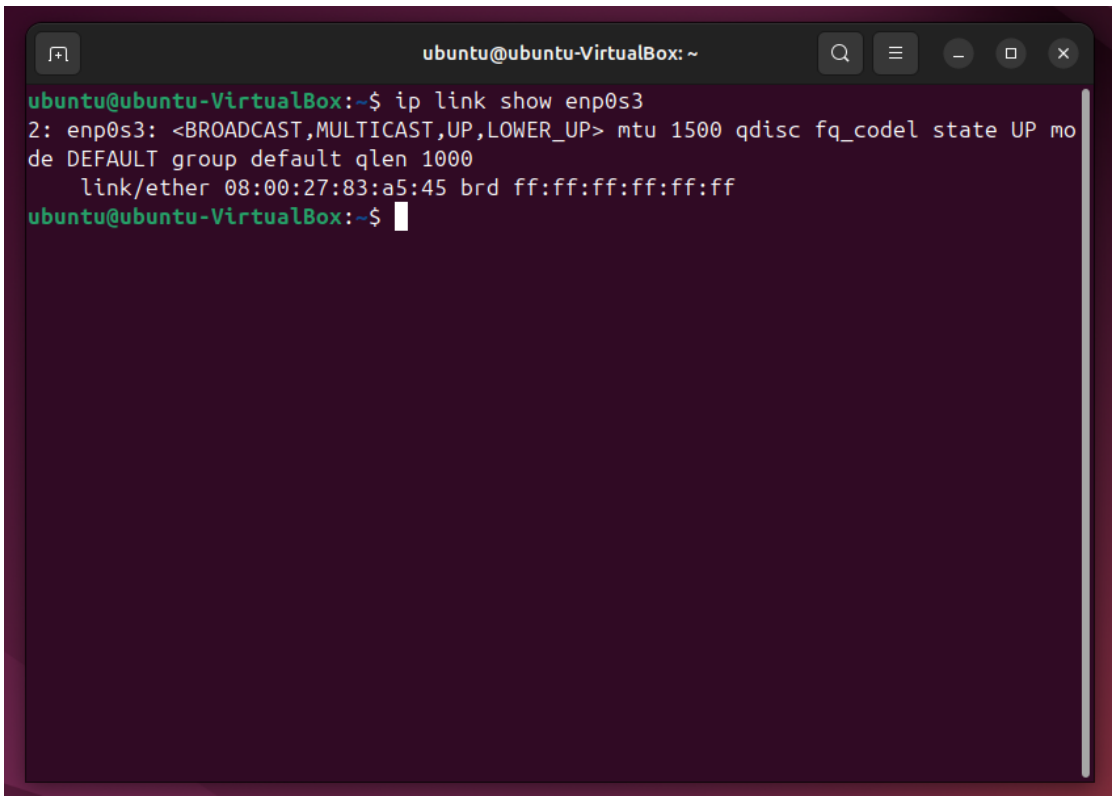


Figure 7: Ubuntu's MAC address

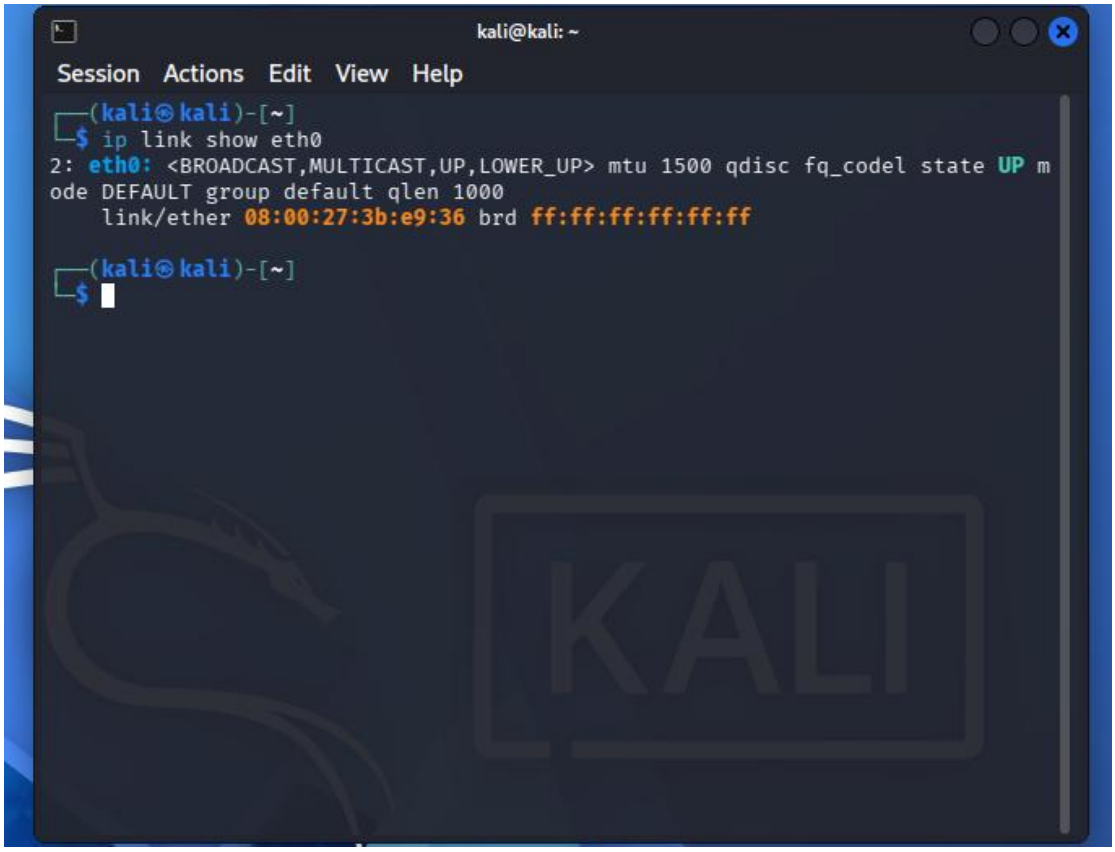


Figure 8: Kali's MAC address

2.3 ARP table before the attack

On the Ubuntu (victim) machine, the ARP table was examined using command *ip neigh show* and it shows the attacker's IP and MAC addresses as well as the gateway's.

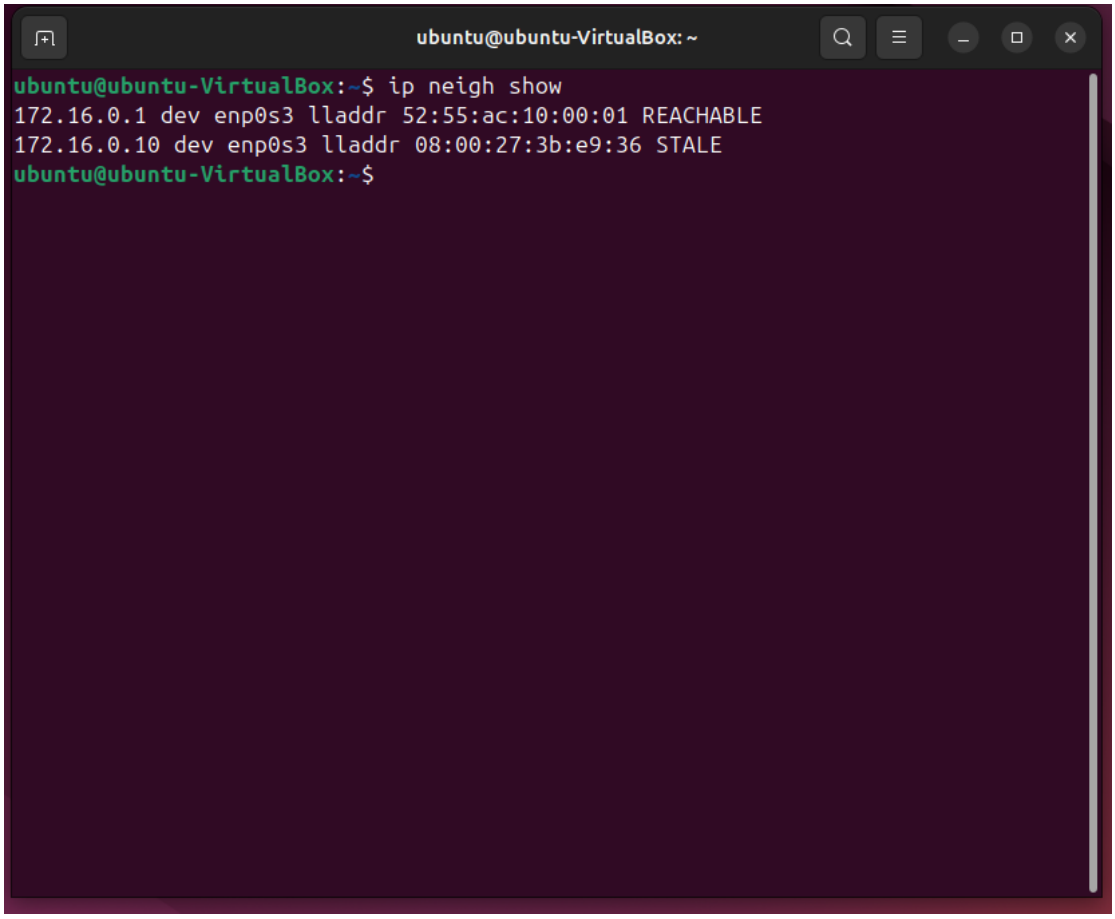


Figure 9: ARP table before the attack on Ubuntu

Similarly, the ARP table on Kali was examined using *arp -a.*

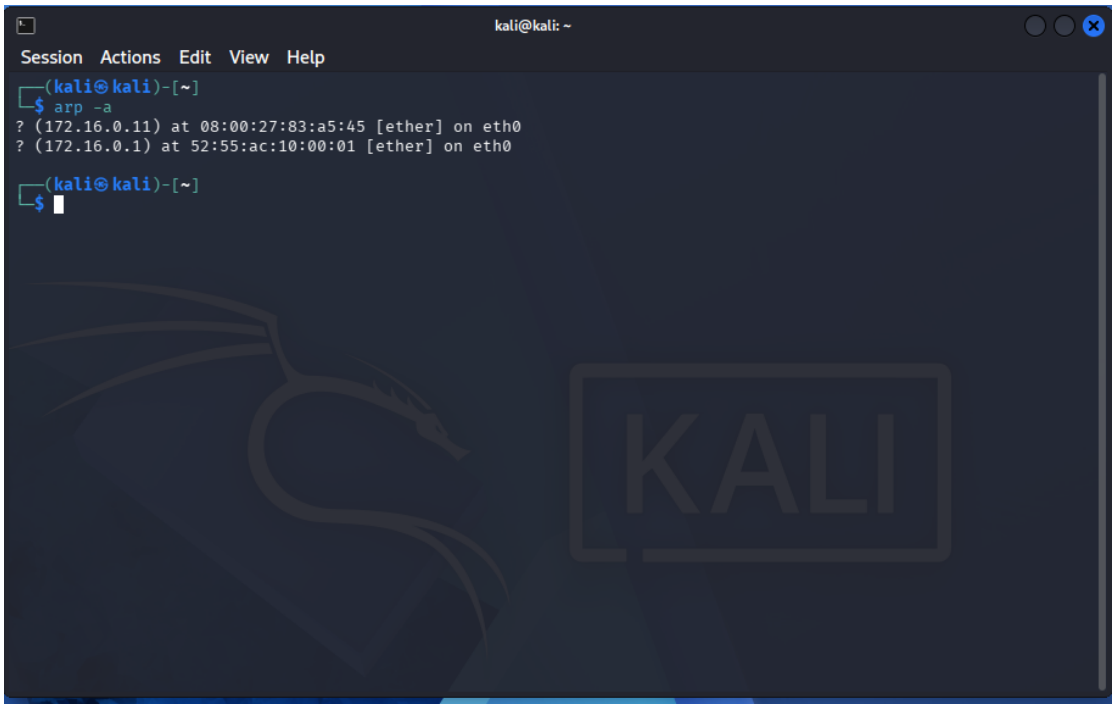


Figure 10: ARP table before the attack on Kali

The following table summarizes the IP-MAC mappings before the attack:

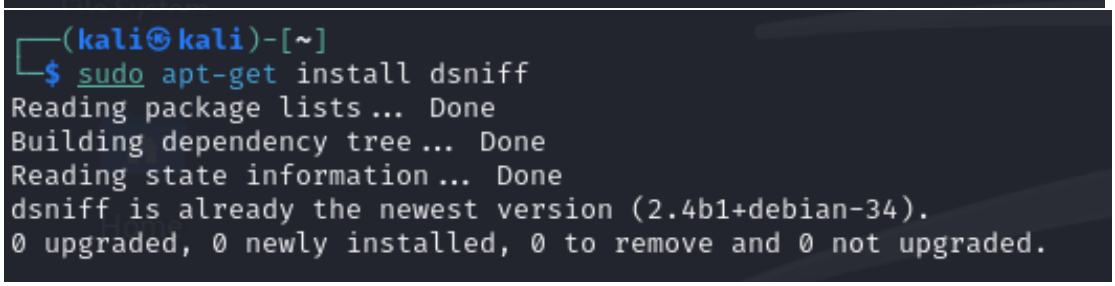
MACHINE	IP ADDRESS	MAC ADDRESS
Kali (attacker)	172.16.0.10/24	08:00:27:3b:e9:36
Ubuntu (victim)	172.16.0.11/24	08:00:27:83:a5:45
Gateway	172.16.0.1	52:55:ac:10:00:01

3. ARP SPOOFING ATTACK

3.1 Enabling IP Forwarding

On the attacker machine (Kali), IP forwarding was enabled so that it could act as a router and forward packets between the victim and the gateway. IP forwarding allows a host to route packets between different networks based on the destination IP address using the system's routing table (Twingate, 2024).





3.2 Executing the ARP Spoofing

Three terminals were opened on Kali.

Terminal 1 – Spoof the victim (Ubuntu) into believing that Kali is the gateway:

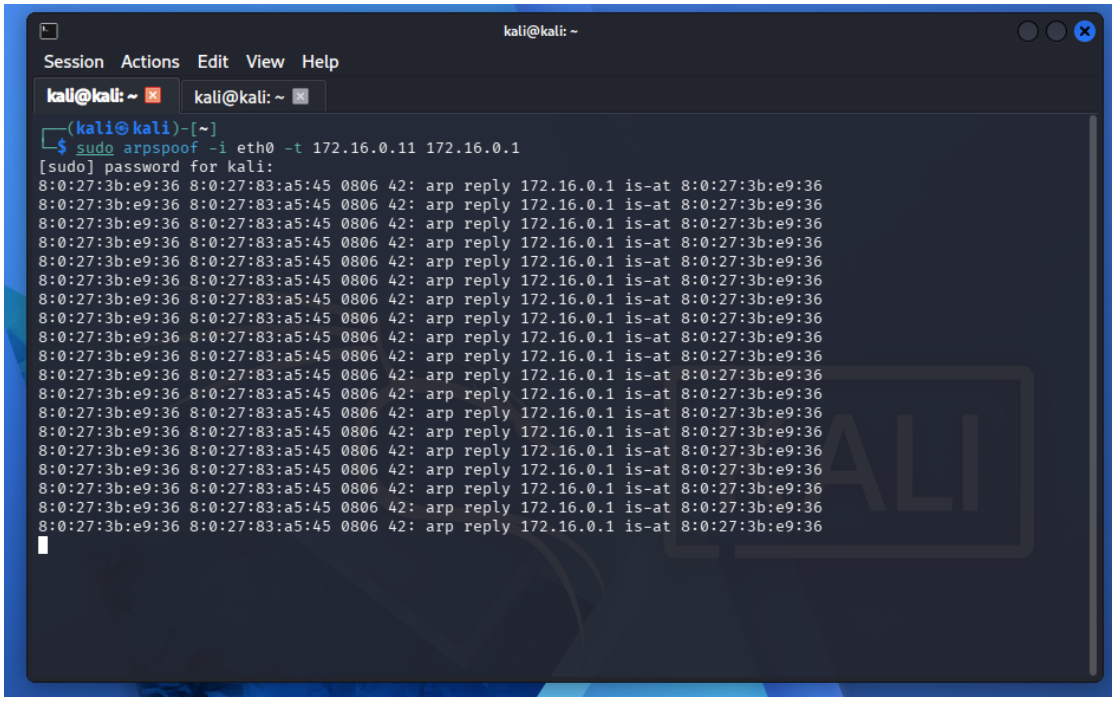


Figure 11: sudo arpspoof -i eth0 -t 172.16.0.11 172.16.0.1

Terminal 2 – Spoof the gateway into believing that Kali is the victim:

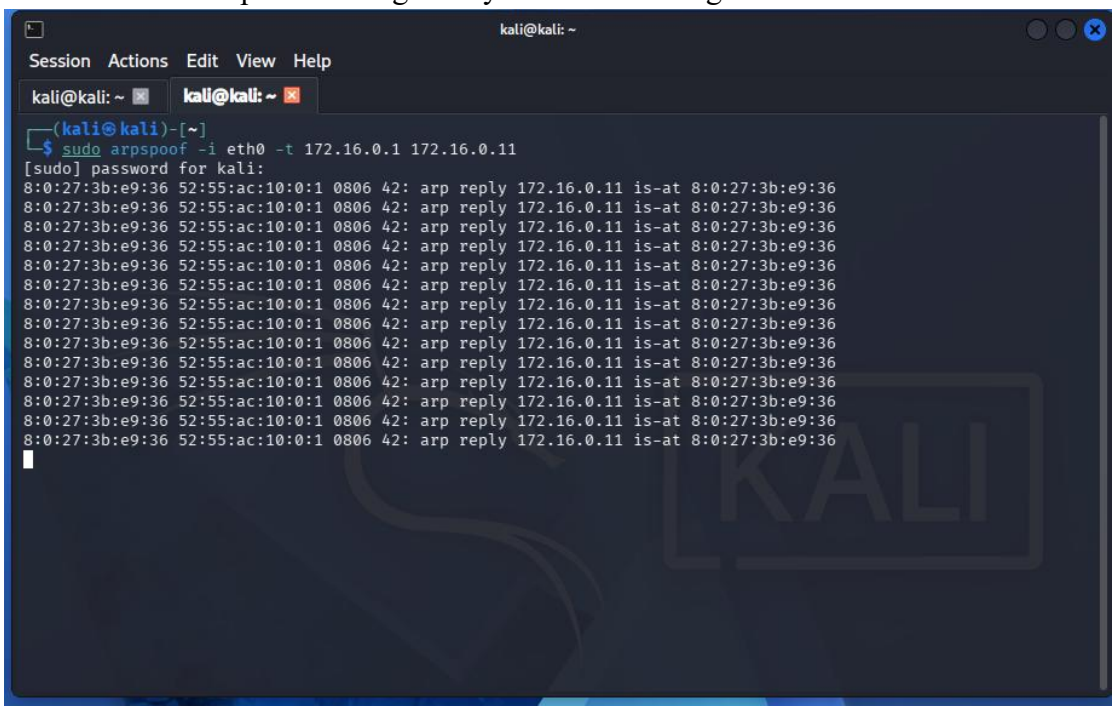


Figure 12: sudo arpspoof -i eth0 -t 172.16.0.1 172.16.0.11

While these two commands were running, a third terminal was used to capture traffic with urlsnarf and Wireshark.

3.3 ARP Table After the Attack

After a few seconds, the ARP table on Ubuntu was checked again. The gateway's MAC address had changed to the MAC address of Kali, confirming the cache poisoning.

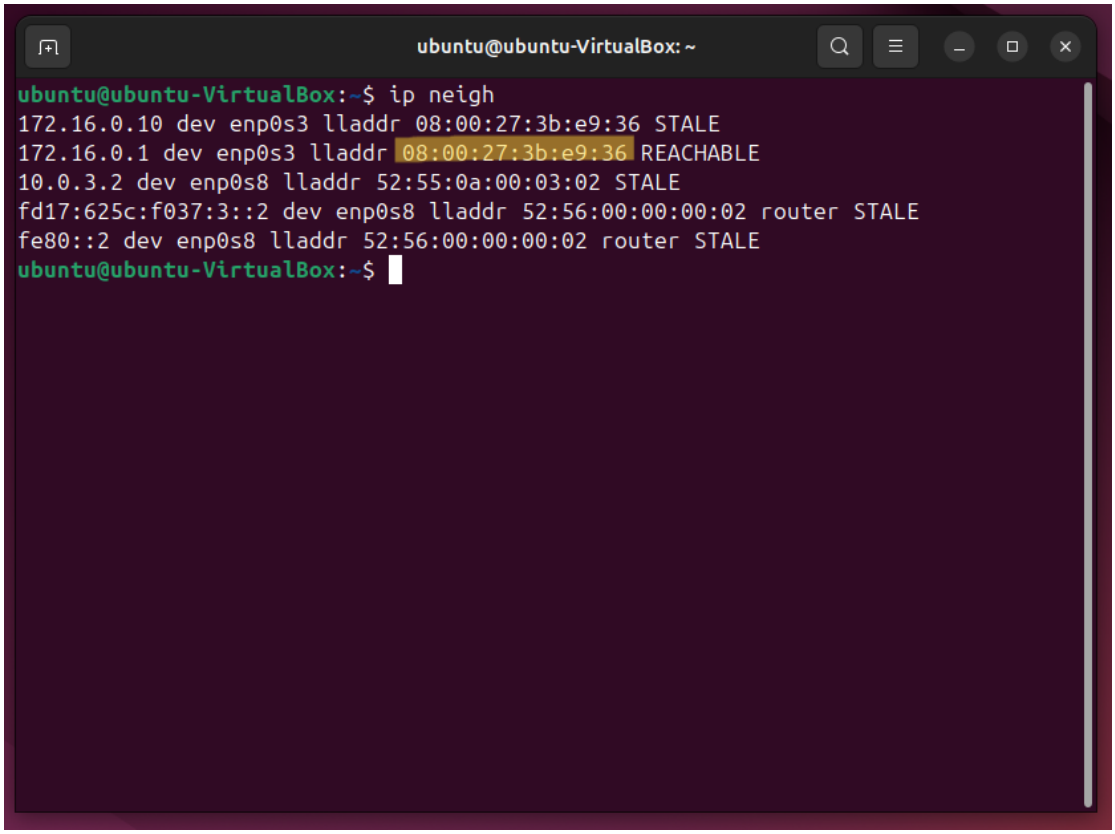


Figure 13: Gateway's MAC has been spoofed to Kali's MAC address

4. Traffic Interception Validation

To validate the MitM position, urlsnarf was launched on Kali.



On Ubuntu, a web browser was opened and an HTTP site was visited. The urlsnarf output displayed the HTTP request from Ubuntu, proving that the traffic was passing through Kali.

Additionally, Wireshark was used to capture ARP packets and to detect duplicate IP warnings, as well as to observe HTTP requests and ICMP traffic passing through the attacker.

During ARP poisoning, Kali sends ARP replies to the victim machine and the router associating Kali's MAC address with the gateway's IP. This creates conflicting ARP entries in devices' ARP tables triggering "duplicate IP warnings".

Wireshark detects and reports these ARP conflicts as "duplicate IP use", which confirms that ARP poisoning is underway.

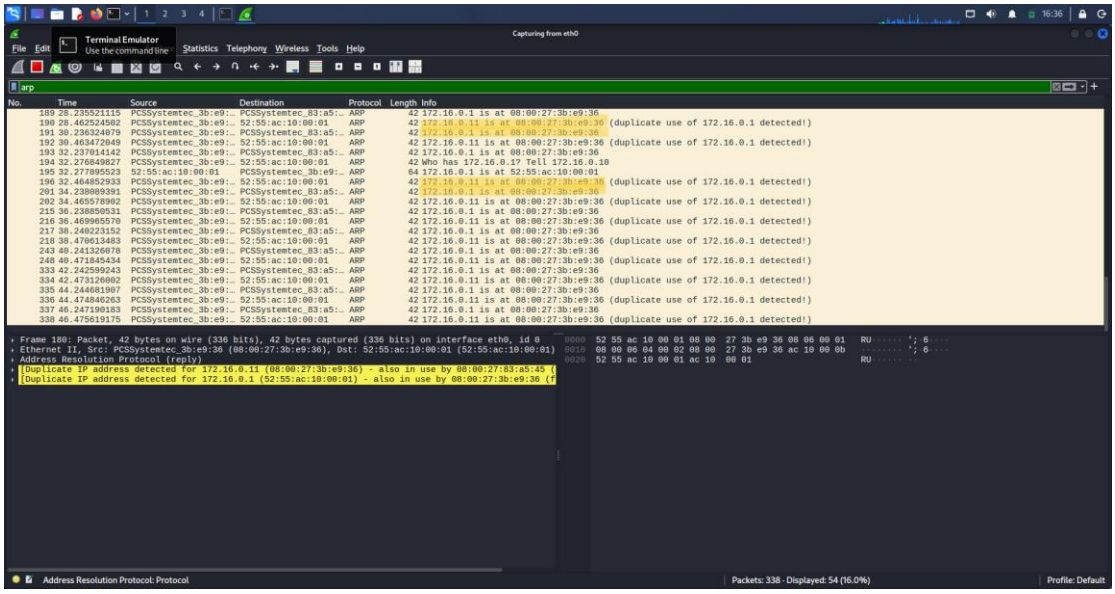


Figure 14: Duplicate uses of the gateway

The filter icmp was applied in Wireshark to observe the echo requests sent from the victim (Ubuntu, IP 172.16.0.11) to an external host (8.8.8.8) and the corresponding echo replies. The captured packets show that the requests first reach the attacker (Kali, MAC 08:00:27:3b:e9:36) before being forwarded to the destination, and the replies follow the reverse path. Additionally, some Destination unreachable (Port unreachable) messages from the gateway (172.16.0.1) to Kali appear, which are likely due to the ongoing ARP cache poisoning causing temporary confusion in the network. The successful round-trip of ICMP packets confirms that IP forwarding is enabled on Kali and that the Man-in-the-Middle position is functional – the victim can still reach the internet, but all traffic passes through the attacker.

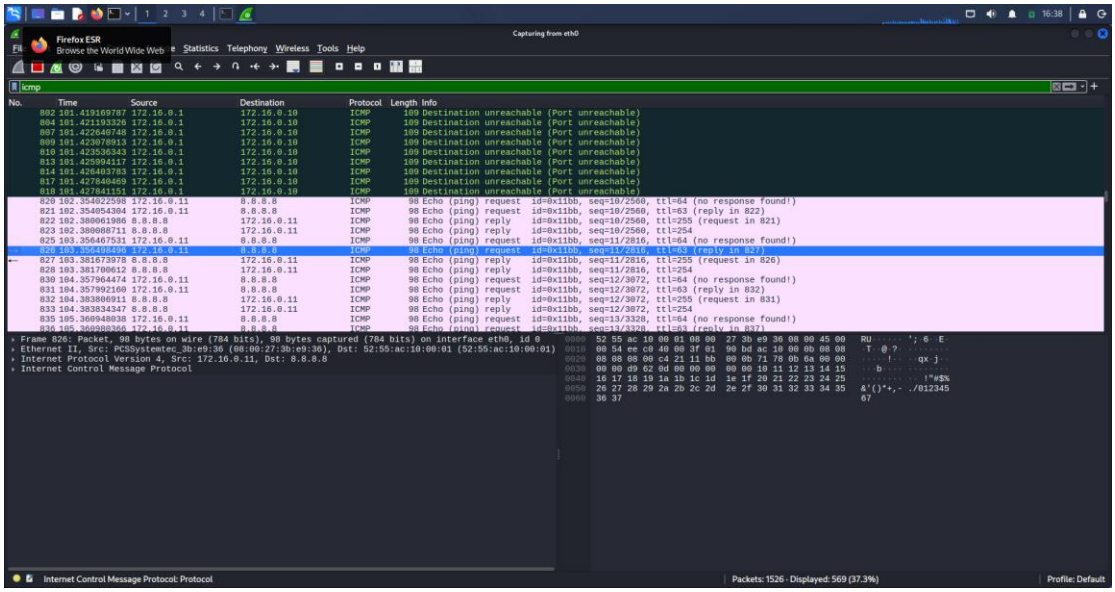


Figure 15: ICMP traffic captured on Kali during the ARP spoofing attack.

With the display filter http.request, Wireshark shows a single HTTP GET request originating from the victim's IP (172.16.0.11) and destined to a remote web server (185.125.190.99 port 80). The Ethernet frame details reveal that the source MAC address belongs to the victim (08:00:27:83:a5:45) while the destination MAC address is that of the attacker (08:00:27:3b:e9:36). This proves that the HTTP traffic from the Ubuntu machine is first sent to the Kali machine – instead of directly to the gateway – because the victim's ARP cache has been poisoned. The attacker is therefore able to intercept and inspect the plaintext HTTP request, confirming the success of the ARP spoofing attack and the MitM position.

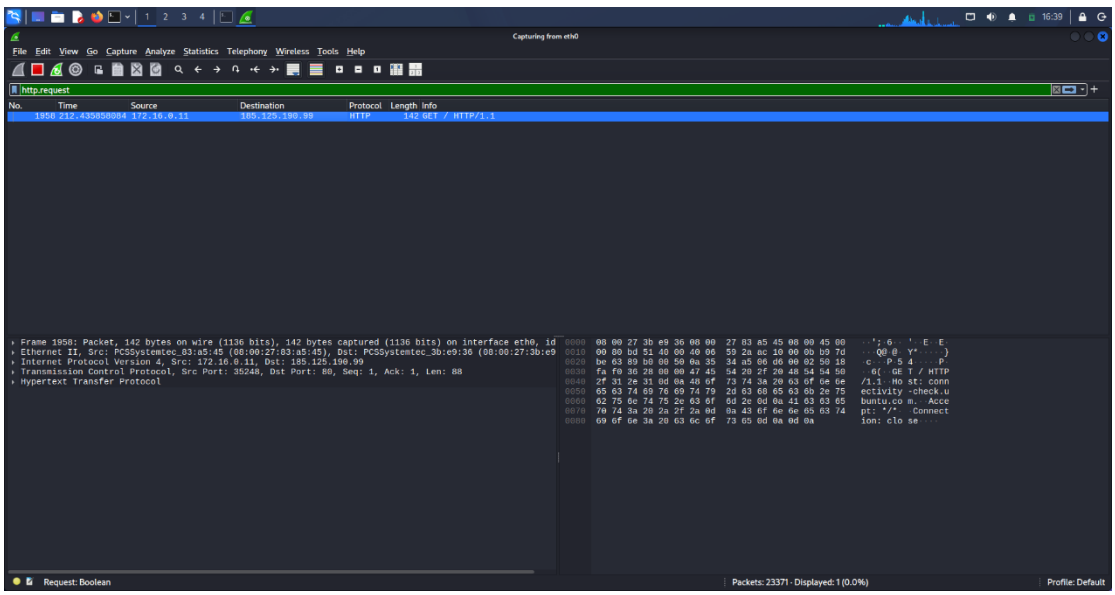


Figure 16: HTTP request from the victim captured on Kali.

5. CONCLUSION

This lab successfully demonstrated an ARP cache poisoning attack in a virtualized environment. By enabling IP forwarding and running two arpspoof instances, the attacker (Kali Linux) was able to place itself in a Man-in-the-Middle position between the victim (Ubuntu) and the gateway. The success of the attack was verified through three independent methods: modification of the ARP table on the victim machine, interception of HTTP traffic using urlsnarf, and detection of duplicate IP warnings in Wireshark.

The results highlight a fundamental weakness of the ARP protocol – its lack of authentication. In real-world networks, such attacks can lead to session hijacking, credential theft, or data manipulation. To mitigate these risks, organizations should implement measures such as Dynamic ARP Inspection (DAI), static ARP entries for critical hosts, and end-to-end encryption (HTTPS, SSH, VPN). Future work could explore more advanced attacks (e.g., ARP poisoning combined with SSL stripping) or defensive mechanisms like ARP spoofing detection tools.

Bibliography

Fortinet. (n.d.). *What is ARP?* Fortinet Cyber Glossary: https://www.fortinet.com/uk/resources/cyberglossary/what-is-arp

Javeed, D., MohammedBadamasi, U., Ndubuisi, C. O., & Asif, M. (2020). *Man in the Middle Attacks: Analysis, Motivation and Prevention*:https://d1wqtxts1xzle7.cloudfront.net/88741461/1Vol8No7-libre.pdf?

Twingate. (2024). *What is IP Forwarding?* Twingate Blog Glossary.:https://www.twingate.com/blog/glossary/ip-forwarding

Yllia, E., & Fejzajb, J. (2021). Man in the Middle: Attack and Protection: https://ceur-ws.org/Vol-2872/short08.pdf